

CHENGDE, China

WMO: 544230

Lat: 40.9726N

Lon: 117.9247E

Elev: 1389

StdP: 13.97

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.9	2.5	-19.6	2.0	9.0	-15.5	2.5	11.5	17.1	20.0	15.1	18.2	2.1	340	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	92.1	69.3	89.4	69.3	86.9	68.8	76.5	84.3	74.8	82.7	73.3	81.1	5.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.5	135.8	81.1	72.6	127.1	79.3	70.9	120.1	78.0	41.3	84.6	39.5	82.9	38.1	81.3	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	13.2	10.7	-5.7	97.4	4.5	3.5	-8.9	100.0	-11.5	102.0	-14.0	104.0	-17.2	106.5
			-6.4	78.7	4.3	2.0	-9.5	80.1	-12.0	81.3	-14.4	82.4	-17.6	83.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	16.8	24.9	38.4	53.2	64.2	71.9	76.4	73.8	63.9	49.9	33.2	19.7	(6)
(7)		DBStd	21.81	6.00	7.29	8.12	7.67	6.28	5.11	4.05	4.42	6.00	6.87	7.75	6.08	(7)
(8)		HDD50	3684	1028	702	370	52	1	0	0	0	1	87	503	940	(8)
(9)		HDD65	6815	1493	1122	825	360	88	8	0	2	89	470	953	1405	(9)
(10)		CDD50	3316	0	0	10	147	442	658	820	739	417	83	0	0	(10)
(11)		CDD65	971	0	0	0	5	64	216	355	275	55	1	0	0	(11)
(12)		CDH74	9215	0	0	3	135	977	2119	3161	2230	572	18	0	0	(12)
(13)		CDH80	3309	0	0	0	19	312	832	1266	754	124	2	0	0	(13)
(14)	Wind	WSAvg	3.4	3.2	3.5	4.4	4.8	4.4	3.4	2.9	2.6	2.6	2.8	2.9	3.0	(14)
(15)	Precipitation	PrecAvg	20.5	0.1	0.2	0.3	0.8	1.7	3.3	5.8	5.1	2.0	0.7	0.2	0.1	(15)
(16)		PrecMax	28.6	1.1	0.9	0.9	2.8	3.7	6.8	11.5	10.8	3.9	2.2	0.9	0.5	(16)
(17)		PrecMin	12.9	0.0	0.0	0.0	0.0	0.5	1.0	1.6	1.1	0.1	0.0	0.0	0.0	(17)
(18)		PrecStd	4.5	0.2	0.2	0.2	0.6	0.9	1.5	2.0	2.8	1.1	0.6	0.2	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	51.1	70.8	82.8	92.7	95.5	97.0	93.2	87.7	76.0	58.8	40.7	(19)
(20)			MCWB	29.0	37.3	48.7	57.8	63.9	65.7	71.5	70.2	67.7	57.3	43.9	31.0	(20)
(21)		2%	DB	34.6	46.2	63.8	78.0	87.1	91.3	92.2	89.4	83.6	71.0	54.4	36.9	(21)
(22)			MCWB	26.4	34.2	45.4	54.3	61.0	66.3	72.1	71.6	64.7	54.7	42.3	29.0	(22)
(23)		5%	DB	31.4	42.7	58.7	73.7	83.2	88.1	89.4	86.9	80.2	66.9	50.5	33.6	(23)
(24)			MCWB	24.3	32.3	42.3	52.5	59.4	65.5	71.9	70.7	63.4	52.7	40.3	26.8	(24)
(25)		10%	DB	28.5	38.9	53.7	69.6	79.6	84.9	86.8	84.4	77.0	63.2	46.4	30.5	(25)
(26)			MCWB	22.4	29.9	39.3	50.3	57.7	65.2	71.6	70.4	62.1	50.4	37.6	24.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.5	39.4	50.7	61.5	67.6	73.2	79.5	78.8	71.5	61.4	46.9	32.2	(27)
(28)			MADB	37.3	48.9	65.8	76.2	83.1	84.9	87.7	86.2	81.4	71.3	54.6	38.8	(28)
(29)		2%	WB	26.9	35.3	46.4	57.4	64.6	70.9	77.3	76.2	68.8	57.9	43.7	29.5	(29)
(30)			MADB	33.7	45.1	61.6	73.0	80.8	81.6	85.4	83.7	77.9	66.2	51.9	36.3	(30)
(31)		5%	WB	24.7	32.6	43.1	54.6	62.5	69.5	75.7	74.5	66.6	55.4	41.3	27.1	(31)
(32)			MADB	31.0	41.9	56.7	68.9	76.8	80.4	83.7	82.0	75.6	63.0	49.0	32.7	(32)
(33)		10%	WB	22.6	30.3	40.2	51.9	60.6	68.0	74.2	72.9	64.4	52.8	38.4	24.9	(33)
(34)			MADB	28.4	38.6	52.7	66.4	73.8	78.2	82.2	80.1	72.8	60.5	46.0	30.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	21.2	22.5	23.3	24.8	25.5	22.2	18.8	19.4	22.9	23.0	20.5	20.0	(35)
(36)			MCDBR	24.6	27.2	30.5	31.6	32.9	28.7	24.2	23.4	27.6	27.6	25.2	23.0	(36)
(37)			MCWBR	18.0	18.1	16.8	14.4	12.6	9.0	7.9	8.4	10.8	13.8	15.0	16.4	(37)
(38)		5% WB	MCDDBR	23.7	25.8	28.5	26.8	26.4	20.7	17.5	17.1	21.0	22.8	23.1	22.0	(38)
(39)			MCWBR	17.6	17.5	16.2	13.2	11.2	8.2	7.3	7.6	9.7	12.8	14.6	15.9	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.345	0.428	0.470	0.550	0.600	0.719	0.703	0.637	0.577	0.521	0.394	0.353	(41)	
(42)		taud	2.166	1.953	1.892	1.763	1.687	1.530	1.602	1.707	1.770	1.823	2.083	2.161	(42)	
(43)		Ebn at Noon	247	237	243	233	225	199	200	209	210	203	224	232	(43)	
(44)		Edh at Noon	33	47	56	69	76	89	82	72	63	53	35	30	(44)	
(45)	All-Sky Solar Radiation	RadAvg	815	1096	1491	1729	1935	1725	1608	1599	1405	1099	821	712	(45)	
(46)		RadStd	50	88	110	105	128	161	122	101	88	84	86	47	(46)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	-3.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page